

DESIGN PROJECT REPORT

DWS-Bench: A Controlled Benchmark for Dynamic World-State Reasoning in Small Language Models

CSE 4610 : Design Project

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List of Abbreviations

SLM	Small Language Model
LLM	Large Language Model
DWS	Dynamic World State
DWS-Bench	Dynamic World-State Benchmark
CO	Course Outcome
RQ	Research Question
E	Entity Count
T	Target-Relevant Update Depth
D	State-Changing Distractor Update Count
N	Narrative Distractor Count
U	Total Canonical Updates
V	Revision Count
L	Rendered Word-Count Proxy
PUT	Object Placement Operation
MOVE	Object Movement Operation
REMOVE	Object Removal Operation
UNDO	History Reversal Operation
REDO	History Reapplication Operation
SPLIT	Object Splitting Operation
MERGE	Object Merging Operation
SWAP	Object Swapping Operation

Abstract

Small Language Models (SLMs) are increasingly deployed where computational efficiency, low latency, and restricted memory are critical. However, their capacity to track dynamic world states across complex state-changing operations remains insufficiently understood. Benchmark evaluations frequently emphasize static knowledge or unconstrained task suites, making it difficult to isolate how performance degrades when entities undergo multi-step state transformations.

This report introduces **DWS-Bench**, a controlled benchmark framework evaluating dynamic world-state reasoning in Small Language Models. DWS-Bench models an environment as a discrete set of entities and locations, generating natural-language narratives from symbolic trajectories spanning eight canonical operations: PUT, MOVE, REMOVE, UNDO, REDO, SPLIT, MERGE, and SWAP. Each instance provides exact, deterministic symbolic ground truth computed independently of its natural-language realization.

Experimental difficulty is controlled via a QuerySpec structural analysis layer. Realized complexity parameters—including temporal update depth (T), entity load (E), state-changing distractors (D), narrative distractors (N), state revision count (V), canonical update volume (U), and rendered word length (L)—are measured for every generated instance. To decouple factor confounding, baseline conditions are matched in length, inspecting residual factor correlations directly.

The investigation is structured around five research questions (RQ1–RQ5) addressing temporal depth, state revision, distractor interference, entity load, and structural operation families. Evaluation proceeds in three systematic phases: (1) generator validation and benchmark freezing; (2) evaluation of three instruction-tuned models (Qwen2.5-0.5B, Qwen2.5-3B, Qwen2.5-7B) across 1,150 frozen instances using direct zero-shot and Chain-of-Thought (CoT) prompting; and (3) an exploratory naturalistic reference audit. The reported model-level results cover RQ1–RQ3 and the structural-operation families in RQ5, while RQ4 remains unevaluated at the model-output level.

Results demonstrate that dynamic state tracking is highly non-monotonic and strongly model- and prompt-dependent. Qwen2.5-3B with CoT achieves the highest final-answer accuracy (49.91%), yet exhibits severe step-wise state trajectory divergence (2.82% step-wise accuracy), revealing a critical gap between final query prediction and true state tracking. The resulting benchmark provides a rigorous, reproducible framework for diagnosing structural failure boundaries in Small Language Models.

Chapter 1

Introduction

1.1 Background

Language models have achieved remarkable capabilities in natural language understanding, generation, and reasoning. A core requirement in interactive and multi-step reasoning environments is the ability to maintain consistent representations of entities as their states evolve over sequences of events. This capability is essential whenever environment states are not restated at every turn but must be inferred from prior observation history.

Dynamic world-state reasoning requires a model to process an ordered narrative of state-changing operations and deduce the state of a target entity at subsequent query points. For instance, an object placed in a room may undergo multiple relocations, removal, or historical reversals. Answering a query regarding its location requires retaining exact state records and applying transitions sequentially. This differs fundamentally from static question answering, where facts can be retrieved from single statements without sequential state tracking.

Prior investigations reveal that entity and state tracking present severe challenges for language models. Kim and Schuster [4] demonstrated that text-only models struggle to preserve entity states without task-specific tuning, experiencing rapid performance decay as operation depth increases. Moreover, they noted that aggregate metrics can be artificially inflated by trivial instances in which target states remain unchanged. Rezaee et al. [14] confirmed that accuracy deteriorates as state update depth grows, with smaller models failing after short operation sequences.

Mechanistic studies clarify these structural limitations. Tang et al. [15] showed that

transformers do not maintain explicit incremental representations of world states across context tokens; instead, relevant information is aggregated in parallel at the final query token. Furthermore, removal operations depend on fragile suppression signals that predict degenerate failure modes. Theoretical work by Merrill et al. [10] proved that both transformers and state-space models without Chain-of-Thought (CoT) or looping are constrained within the circuit complexity class TC^0 , which provably excludes general state tracking such as permutation composition. These insights underscore the necessity of benchmarks that systematically control structural complexity rather than relying on unconstrained natural text.

1.2 Motivation and Scope

Small Language Models (SLMs)—defined here operationally as compact instruction-tuned models under 10 billion parameters—offer compelling advantages for edge computing, low-latency applications, and privacy-preserving deployments. However, their reduced parameter capacity makes understanding their state-tracking boundaries paramount.

A key methodological challenge in evaluating dynamic reasoning is factor coupling: increasing target update depth naturally lengthens contexts, adding entities introduces distracting references, and inserting unrelated state changes increases noise. Without decoupling these dimensions, observed performance degradation cannot be cleanly attributed to a specific structural factor.

DWS-Bench resolves this by decoupling symbolic state generation from natural-language rendering. The symbolic simulator computes exact state transition histories and validates experimental contract compliance before rendering narratives. The benchmark covers eight operations (PUT, MOVE, REMOVE, UNDO, REDO, SPLIT, MERGE, SWAP) across multiple trajectory families. For every instance, seven realized complexity variables are recorded:

- E : Entity count;
- T : Target-relevant update depth;
- D : State-changing distractor update count;
- N : Narrative distractor count;
- V : Revision count;
- U : Total canonical updates;
- L : Rendered word-count proxy.

State-changing distractor updates contribute to U because they are canonical operations modifying the symbolic world. Narrative distractors do not change symbolic state and therefore do not contribute to U .

1.3 Problem Statement

Small Language Models struggle to preserve accurate world-state representations as event sequence complexity increases. Existing benchmarks frequently fail to isolate temporal depth, interference, and history revision, leaving open whether performance drops stem from sequence length, distractor load, or structural state operations.

This thesis addresses the central problem:

How can the dynamic world-state reasoning capability of Small Language Models be systematically evaluated using a controlled benchmark that isolates temporal depth, state revision, distractor interference, entity load, and structural operation families?

1.4 Research Questions

- **RQ1 – Temporal Depth:** How does increasing target-relevant update depth (T) affect accuracy, and where do failure boundaries occur?
- **RQ2 – State Revision:** How does overwriting state history (re-visiting states or applying UNDO/REDO) impact state tracking performance?
- **RQ3 – Distractor Interference:** How do state-changing distractors (D) versus narrative distractors (N) impact accuracy under length-matched controls?
- **RQ4 – Entity Load:** Does increasing entity count (E) degrade target state tracking accuracy under fixed target update depth?
- **RQ5 – Structural Operation Families:** How do complex structural operations (SPLIT, MERGE, SWAP, UNDO, UNDO-REDO) affect reasoning accuracy and error mechanisms?

1.5 Research Objectives

The objectives of this research are:

1. To design and implement a deterministic simulator capable of generating controlled dynamic world-state trajectories.
2. To implement an eight-operation vocabulary and trajectory-family taxonomy for dynamic world-state reasoning.
3. To develop a QuerySpec-based structural analysis layer that selects queries satisfying explicit difficulty requirements.
4. To develop a validator that verifies symbolic state transitions and experimental-contract compliance before rendering.
5. To measure realized complexity factors (E, T, D, N, V, U, L) and inspect residual correlations among them.
6. To generate canonical intermediate states supporting trajectory-level evaluation and future counterfactual probing.
7. To freeze validated benchmark datasets before the main model evaluation.
8. To evaluate three instruction-tuned Small Language Models (Qwen2.5-0.5B, 3B, 7B) using final-answer and step-wise accuracy.
9. To analyze performance degradation and error dynamics as dynamic-state complexity increases.
10. To audit whether controlled findings remain qualitatively relevant in naturalistic procedural narratives.

1.6 Contributions

The contributions of this thesis are:

1. A controlled benchmark framework for sequential dynamic world-state reasoning in Small Language Models.
2. A deterministic symbolic trajectory generator and validator providing exact canonical ground truth.
3. A QuerySpec-based generation methodology making structural difficulty explicit and measurable.
4. An eight-operation vocabulary and trajectory-family taxonomy for dynamic world-state reasoning.
5. A decoupled factor-measurement scheme recording realized complexity factors and inspecting residual associations.
6. Step-wise canonical gold trajectories supporting trajectory-level evaluation and future counterfactual probing.
7. A standardized evaluation protocol for three instruction-tuned Small Language Models.

8. A performance-degradation and error-dynamics analysis framework.
9. A naturalistic reference audit for assessing qualitative transfer of controlled findings.

Chapter 2

Course Outcome Alignment and Project Compliance

This chapter documents how the DWS-Bench project satisfies the required Course Outcomes through its research, implementation, evaluation, and project-management activities.

Table 2.1: Mapping of DWS-Bench project artifacts to Course Outcomes.

CO	Course Outcome	DWS-Bench Implementation / Artifact
CO1	Research / State-of-the-Art	Review of SLM reasoning, entity tracking literature, theoretical boundaries (TC ⁰), and development of a controlled state benchmark.
CO2	Teamwork / Project Management	Technical Contribution Matrix, PR records (2 per member), commit logs, weekly meeting summaries, and repository metrics.
CO3	Ethics / Environment	Ground-truth separation preventing hallucinated labels, transparent synthetic dataset generation, and evaluation energy analysis.
CO4	Communication	Structured thesis report, empirical breakdown across RQ1–RQ5, formal presentation slides, TikZ diagrams, and result tables.
CO5	Future Work	Modular architecture enabling expanded query types, relational states, multi-hop reasoning, and larger model evaluation.

2.1 Project Management and Compliance Artifacts

The following project artifacts are maintained as evidence of research and implementation progress:

1. Technical Contribution Matrix;
2. Two distinct PRs per team member;
3. GitHub contributor statistics;
4. Commit-history evidence;
5. Weekly meeting summaries;
6. Complete benchmark source code;
7. Phase 1 generator-validation results;
8. Completed model-level evaluation results for reported conditions;
9. Formal presentation.

2.2 Implementation Milestones

The research was executed across three main phases as summarized in Table 2.2.

Table 2.2: DWS-Bench research phases.

Phase	Main Activity	Purpose
Phase 1	Generator validation & benchmark freezing	Measure realized factor distributions, verify QuerySpec constraints, freeze 1,150 benchmark instances.
Phase 2	SLM evaluation & error analysis	Evaluate Qwen2.5 models (0.5B, 3B, 7B) on final and step-wise accuracy across zero-shot and CoT prompting.
Phase 3	Naturalistic reference audit	Examine qualitative transfer to unconstrained procedural narratives.

Chapter 3

Related Work

3.1 Entity Tracking in Language Models

Entity tracking requires maintaining state representations as entities undergo sequential operations. Kim and Schuster [4] evaluated text-only language models on entity tracking, finding substantial error rates as operation depth grew. Crucially, their re-analysis of probing datasets from Li et al. [7] revealed that aggregate accuracy figures were frequently skewed by trivial non-updating instances, demonstrating the necessity of rigorous ground-truth validation. Tang et al. [15] provided mechanistic insights showing that transformers do not maintain step-by-step state representations, instead aggregating information at final query tokens, with specific operations (e.g., REMOVE) relying on fragile suppression circuits.

3.2 State Tracking and Dynamic Reasoning

Dynamic state tracking formalizes world state S_t at step t , updated via operation o_t according to $S_{t+1} = F(S_t, o_t)$. Rezaee et al. [14] evaluated state tracking across task domains including LinearWorld, HandSwap, and Lights. They showed that earlier models fail rapidly under increased update depth, whereas stronger models equipped with CoT maintain accuracy over longer sequences.

3.3 Theoretical Limits and Operation Mechanisms

Theoretical bounds established by Merrill et al. [10] prove that transformers and state-space models belong to circuit complexity class TC^0 , preventing exact state tracking

without CoT scratchpads or looping. Low-depth shortcuts explored by Liu et al. [9] show models often exploit structural heuristics. Mechanistic studies by Li et al. [6], Prakash et al. [12], Feng and Steinhardt [3], Dai et al. [1], and Prakash et al. [13] illustrate that entity binding and lookback mechanisms drive internal state representations.

3.4 Scale, Naturalism, and State Revision

The relationship between model scale and tracking remains actively debated. While Kim et al. [5] argued multi-billion parameter scale with code pretraining is necessary, Drozd and Heilbron [2] observed implicit tracking in 410M parameter models on naturalistic text. Internal state representations have been decoded by Li et al. [8] and Nanda et al. [11], though Vafa et al. [16] caution that generative world models often lack global coherence. State revision and overwriting, central to RQ2, connect to belief revision studies (Wilie et al. [19]). Distractor interference work (Yang et al. [20], Unable et al. [17], Distractor et al. [18]) highlights the necessity of length-matched experimental design.

3.5 Research Gap

DWS-Bench targets five critical gaps:

1. Insufficient characterization of Small Language Models (<10B parameters) under dynamic state tracking.
2. Confounding of temporal depth, distractors, and narrative length in prior benchmarks [4], [14].
3. Lack of explicit isolation of state revision and history reversal operations.
4. Absence of explicit analysis targeting the transition from stable performance to failure-onset behavior.
5. Reliance on natural-language realization for ground truth rather than deterministic symbolic simulation.

Chapter 4

Problem Definition and Experimental Contract

4.1 Formal World Model and Operation Vocabulary

A benchmark world is defined by a finite entity set $\mathcal{E} = \{e_1, e_2, \dots, e_n\}$ and location set $\mathcal{L} = \{l_1, l_2, \dots, l_m\}$. World state S_t evolves over trajectory $\tau = (o_1, o_2, \dots, o_U)$ via $S_{t+1} = F(S_t, o_t)$.

DWS-Bench implements eight canonical operations:

$$\mathcal{O} = \{\text{PUT, MOVE, REMOVE, UNDO, REDO, SPLIT, MERGE, SWAP}\} \quad (4.1)$$

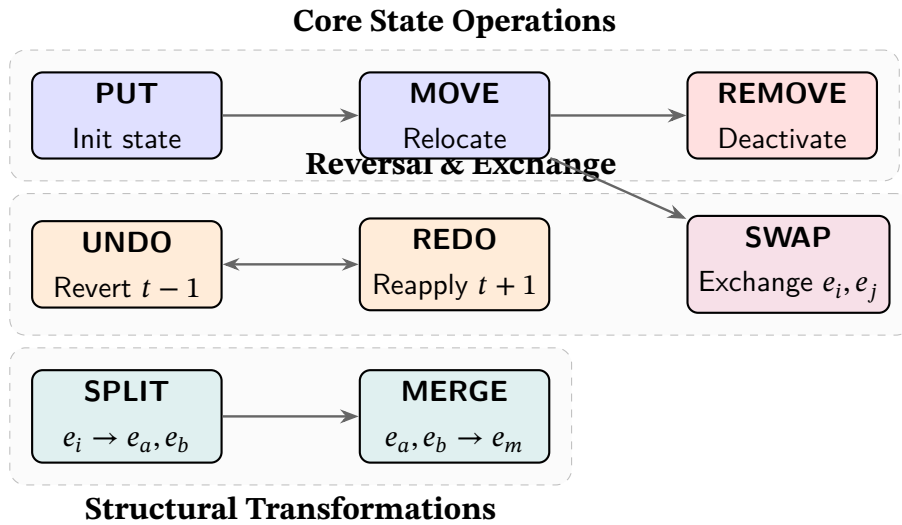


Figure 4.1: DWS-Bench eight-operation vocabulary taxonomy.

4.2 Experimental Factors and Structural Control

Every instance records seven realized factors (Table 4.1). Baseline distractor conditions use length-matching controls (± 3 words) using narrative distractors (N) to isolate symbolic updates (D) from raw token length (L).

Table 4.1: DWS-Bench recorded complexity factors.

Factor	Description	Experimental Role
E	Entity count	World state entity load
T	Target-relevant update depth	Temporal dependency depth
D	State-changing distractor updates	Symbolic state interference
N	Narrative-level distractors	Surface text length equalizer
V	State revision count	History overwriting pressure
U	Total canonical updates	Overall trajectory size
L	Rendered word count proxy	Context length diagnostic

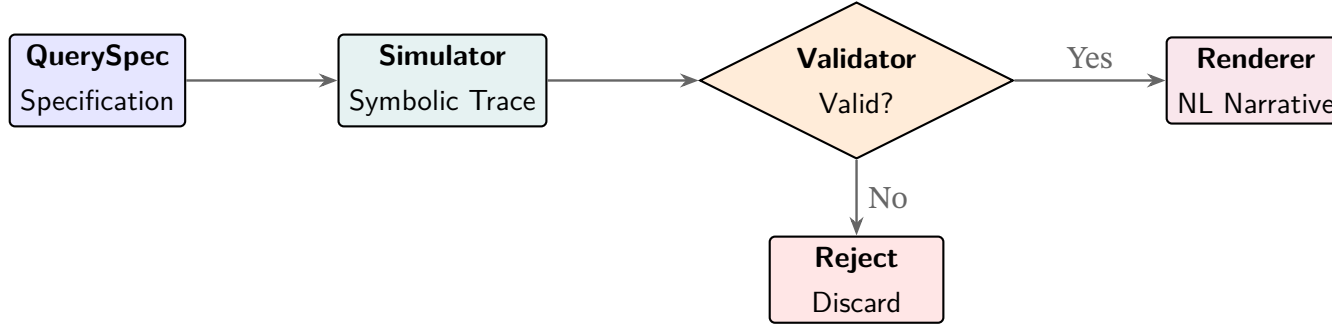


Figure 4.2: DWS-Bench trajectory validation flow.

Ground truth labels are strictly computed from symbolic simulation S_U :

$$\text{GroundTruth} = G(S_U, q) \quad (4.2)$$

The natural-language renderer cannot alter symbolic gold labels, preventing ground-truth drift.

Chapter 5

Methodology

5.1 System Architecture

The DWS-Bench workflow separates symbolic trajectory construction, validation, narrative rendering, and model evaluation into a modular pipeline (Figure 5.1).

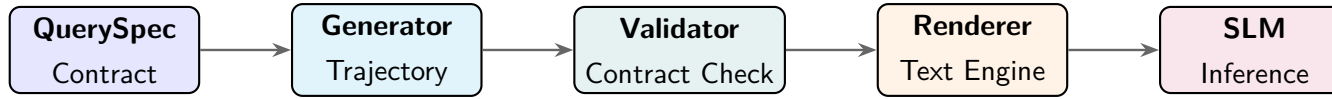


Figure 5.1: DWS-Bench system architecture pipeline.

DWS-Bench Decoupled Factor Taxonomy



Figure 5.2: DWS-Bench factor taxonomy separating state complexity from surface noise.

5.2 Evaluation Protocols

Evaluating models across three Qwen2.5 parameter variants (0.5B, 3B, 7B) under two prompting regimes (Direct Zero-Shot and Chain-of-Thought).

- **Final-Answer Accuracy:** String matching normalized state outputs against S_U .
- **Step-Wise Accuracy:** Evaluating inferred states $\hat{S}_t$ against symbolic intermediate states S_t at every target update step $t \in \{1, \dots, T\}$.

5.3 Counterfactual Probe Support

The repository contains support for constructing counterfactual probes by changing a controlled operation or state transition while preserving the surrounding context as closely as the generator permits. No populated counterfactual probe results are included in the current evaluation artifacts, so this capability is treated as infrastructure for future experiments rather than as an empirical finding in this thesis.

Chapter 6

Implementation

6.1 Software Architecture and Status

The benchmark repository is structured into decoupled Python modules (Table 6.1).

Table 6.1: DWS-Bench implementation architecture.

Module	Responsibility
Contract / QuerySpec	Defines condition rules and structural constraints.
Simulator	Maintains exact symbolic state transitions and history.
Generator	Constructs candidate trajectories satisfying QuerySpecs.
Validator	Enforces trajectory admissibility and contract rules.
Renderer	Translates symbolic steps into natural language.
Evaluator	Runs SLM inference and computes step/final metrics.

Table 6.2 summarizes current implementation status.

Table 6.2: Implementation status summary.

Component	Status	Verification Evidence
Symbolic Simulator	Completed	Unit tests & state assertions
8-Operation Vocabulary	Completed	Synthetic state unit tests
QuerySpec Layer	Completed	Specification contract tests
Validator & Sampler	Completed	Admissibility suite
Phase 1 Validation	Completed	Factor distribution analysis
Phase 2 Model Suite	Completed	1,150 frozen instances evaluated
Phase 3 Audit	Pilot	Naturalistic reference corpus

Chapter 7

Results and Discussion

7.1 Phase 1: Generator Validation

Phase 1 verified that generated instances meet QuerySpec bounds prior to freezing (Table 7.1).

Table 7.1: Phase 1 factor validation summary on frozen benchmark.

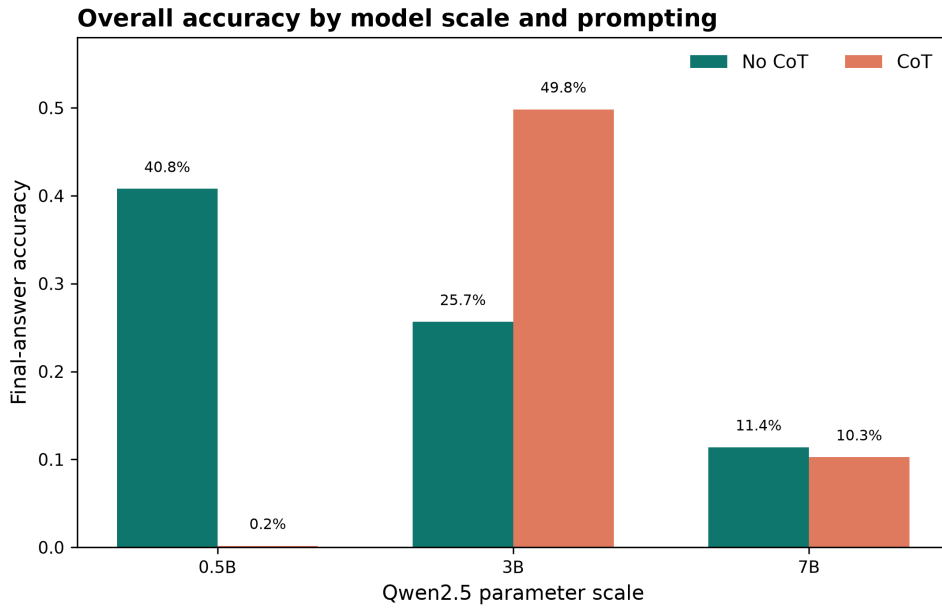
Factor	Requested Range	Realized Range	Validation Status
E	1–5	1–5	Verified across benchmark
T	2–16	2–16	Realized in depth sweep
D	0–16	0–16	Realized in distractor sweep
N	{0, 4}	{0, 4}	Matched length control
L	< 600 words	32–312 words	Word-count proxy verified

7.2 Phase 2: Benchmark Model Performance

Table 7.2 details overall benchmark performance across all 1,150 frozen instances.

Table 7.2: Overall benchmark performance across evaluated models.

Model	Prompting	Final Accuracy	Step-wise Accuracy	Instances
Qwen2.5-0.5B	Direct	40.87%	25.39%	1150
Qwen2.5-0.5B	CoT	0.17%	–	1150
Qwen2.5-3B	Direct	25.65%	24.07%	1150
Qwen2.5-3B	CoT	49.91%	2.82%	1150
Qwen2.5-7B	Direct	11.39%	–	1150
Qwen2.5-7B	CoT	10.52%	23.99%	1150

**Figure 7.1:** Overall final-answer accuracy across model scales and prompting conditions.

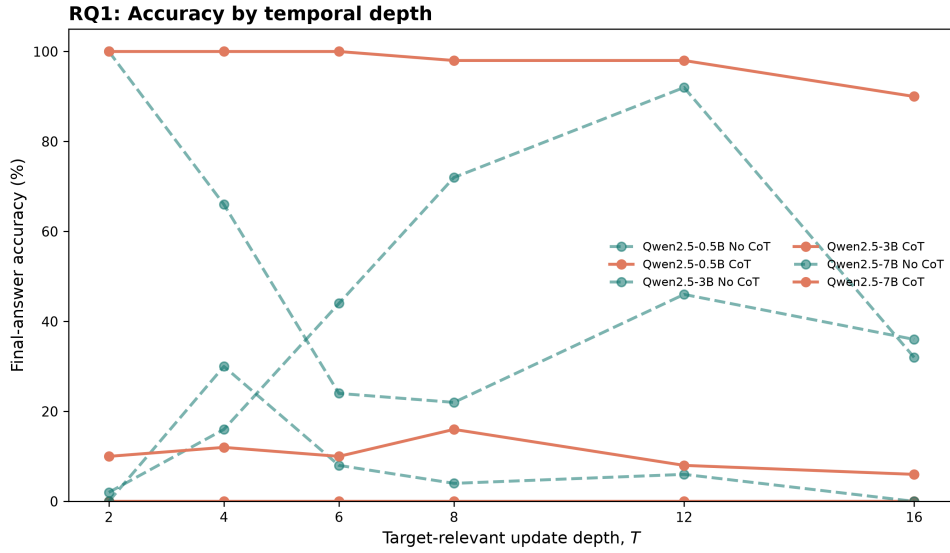
The highest overall accuracy (49.91%) was achieved by Qwen2.5-3B CoT. However, its step-wise accuracy was only 2.82%, demonstrating that final answer correctness frequently occurred despite severe intermediate state tracking breakdown. Qwen2.5-0.5B CoT collapsed to 0.17%, showing CoT prompting can severely degrade performance in ultra-small models.

7.3 RQ1: Temporal Depth Results

Table 7.3 and Figure 7.2 present final-answer accuracy as update depth T increases from 2 to 16.

Table 7.3: Final-answer accuracy by target update depth (T) (RQ1).

Model & Prompt	$T = 2$	$T = 4$	$T = 6$	$T = 8$	$T = 12$	$T = 16$
Qwen2.5-0.5B Direct	100.0%	66.0%	24.0%	22.0%	46.0%	36.0%
Qwen2.5-0.5B CoT	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Qwen2.5-3B Direct	85.0%	88.0%	84.0%	88.0%	84.0%	78.0%
Qwen2.5-3B CoT	100.0%	100.0%	100.0%	98.0%	98.0%	90.0%
Qwen2.5-7B Direct	6.0%	8.0%	14.0%	10.0%	12.0%	8.0%
Qwen2.5-7B CoT	10.0%	12.0%	10.0%	16.0%	8.0%	6.0%

**Figure 7.2:** RQ1 final accuracy across temporal update depths (T).

7.4 RQ2: State Revision Results

Table 7.4 reports performance under state overwriting and revision (V).

Table 7.4: Final accuracy on state revision sweep (RQ2).

Model & Prompt	$T = 4$	$T = 8$	$T = 12$	$T = 16$
Qwen2.5-0.5B Direct	48.0%	40.0%	40.0%	30.0%
Qwen2.5-0.5B CoT	0.0%	0.0%	0.0%	0.0%
Qwen2.5-3B Direct	6.0%	10.0%	8.0%	4.0%
Qwen2.5-3B CoT	6.0%	76.0%	2.0%	2.0%
Qwen2.5-7B Direct	4.0%	0.0%	4.0%	0.0%
Qwen2.5-7B CoT	4.0%	14.0%	0.0%	0.0%

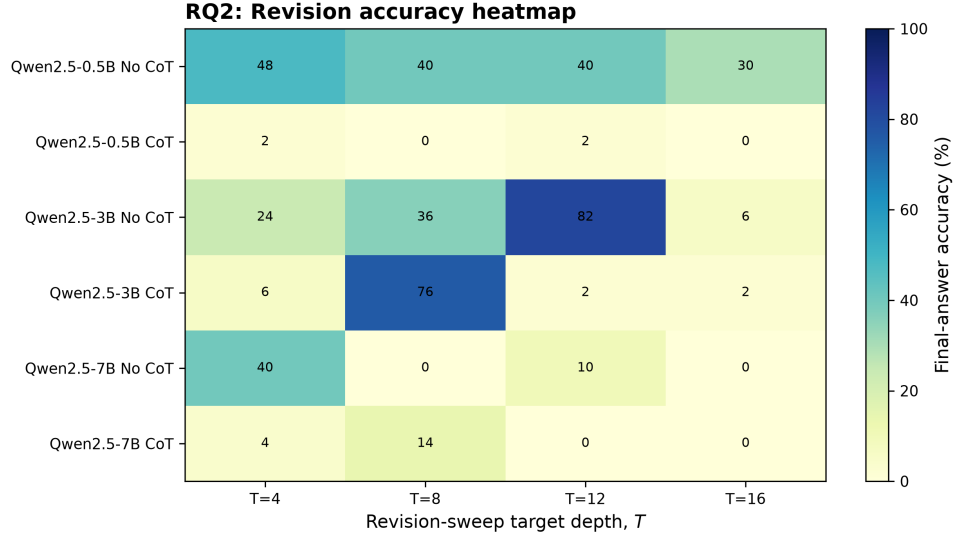


Figure 7.3: RQ2 accuracy across state revision conditions.

7.5 RQ3: Distractor Interference Results

Table 7.5 details performance under state-changing distractors (D).

Table 7.5: Final accuracy on distractor sweep (RQ3).

Model & Prompt	$D = 4$	$D = 8$	$D = 16$
Qwen2.5-0.5B Direct	29.0%	38.0%	10.0%
Qwen2.5-0.5B CoT	0.0%	0.0%	0.0%
Qwen2.5-3B Direct	18.0%	16.0%	16.0%
Qwen2.5-3B CoT	18.7%	16.0%	16.0%
Qwen2.5-7B Direct	0.0%	0.0%	0.0%
Qwen2.5-7B CoT	7.0%	36.0%	52.0%

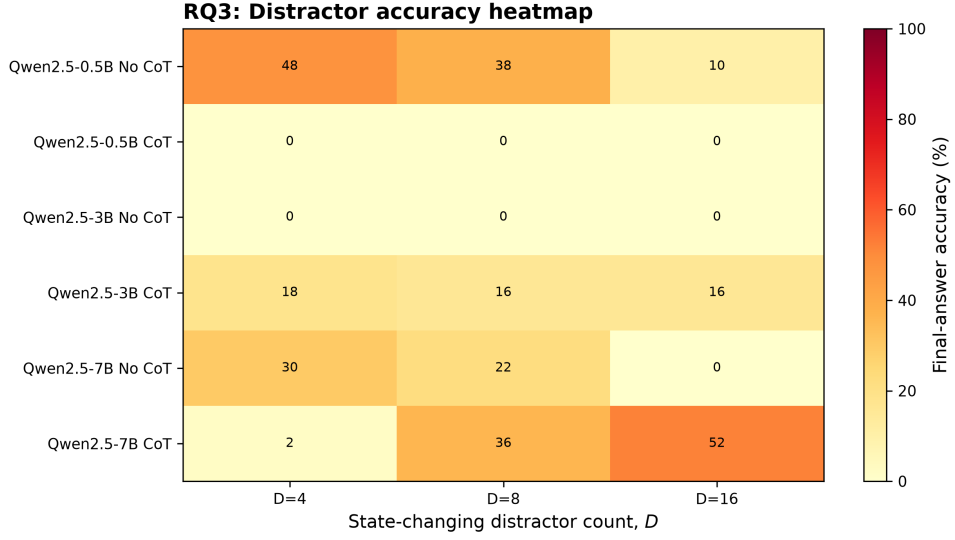


Figure 7.4: RQ3 accuracy across state-changing distractor counts.

7.6 RQ4: Entity Load Status

Entity load sweeps ($E \in \{2, 3, 4, 5\}$) were generated in the benchmark, but evaluation summaries remain pending at the model output level.

7.7 RQ5: Structural Operation Pilot Results

Table 7.6 presents accuracy across five complex operation families (250 instances). Realized factors must be reported: the `split_chain` records realize $T_{\text{actual}} = 7$ and $D_{\text{actual}} = 1$, while the available records for the other four pilot families realize $T_{\text{actual}} = 8$ in the frozen data. Because the `split` family has a different realized target depth, these family comparisons should be interpreted descriptively rather than as perfectly matched causal contrasts.

Table 7.6: Final accuracy on structural operation families (RQ5).

Model & Prompt	Split	Merge	Swap	Undo	Undo-Redo
Qwen2.5-0.5B Direct	12.0%	68.0%	56.0%	60.0%	68.0%
Qwen2.5-0.5B CoT	0.0%	0.0%	0.0%	0.0%	0.0%
Qwen2.5-3B Direct	8.0%	18.0%	20.0%	58.0%	80.0%
Qwen2.5-3B CoT	66.0%	92.0%	18.0%	92.0%	64.0%
Qwen2.5-7B Direct	0.0%	0.0%	8.0%	0.0%	0.0%
Qwen2.5-7B CoT	8.0%	6.0%	6.0%	2.0%	4.0%

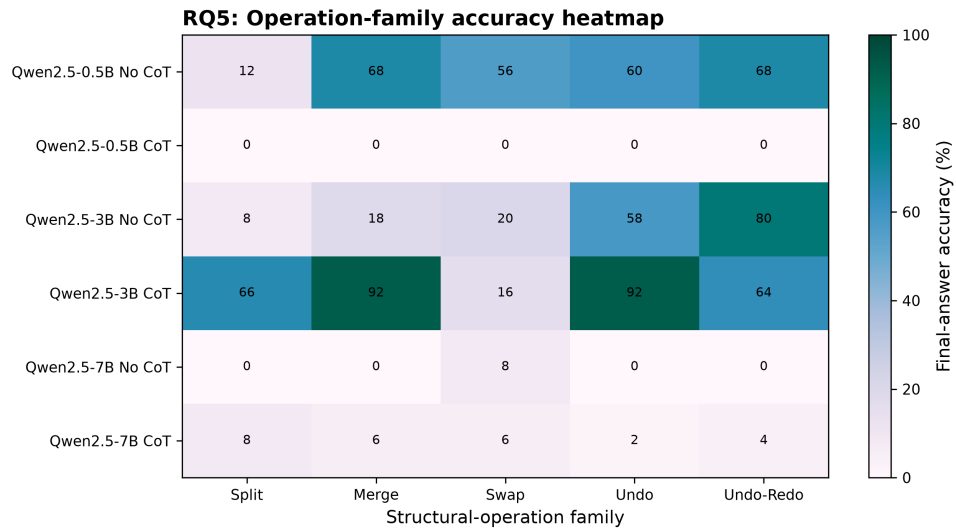


Figure 7.5: RQ5 accuracy across structural operation families.

7.8 Step-Wise vs Final-Answer Error Dynamics

Figure 7.6 compares final answer versus step-wise trajectory accuracy.

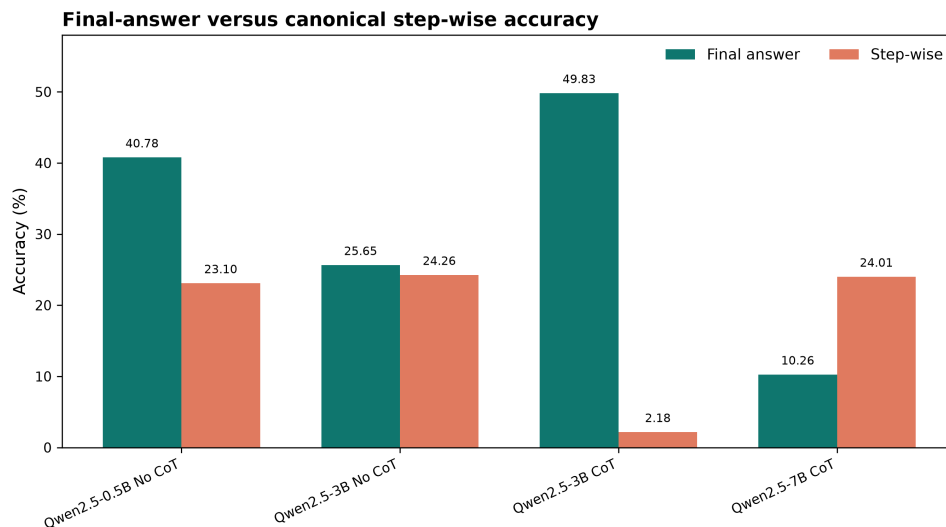


Figure 7.6: Final-answer versus step-wise accuracy comparison.

The stark discrepancy for Qwen2.5-3B CoT (49.91% final vs. 2.82% step-wise) indicates that while CoT helps retrieve final answers, it fails to maintain intermediate state accuracy.

7.9 Research Question Findings Summary

Table 7.7 synthesizes key empirical conclusions across research questions.

Table 7.7: Summary of empirical findings across RQs.

RQ	Empirical Summary
RQ1	Temporal depth causes non-monotonic degradation; 3B CoT maintains >90% accuracy through $T = 16$, whereas 0.5B Direct drops to 24% at $T = 6$.
RQ2	State revision induces severe failure; 3B CoT drops to 2.0% at $T = 12, 16$.
RQ3	Distractor updates degrade small model accuracy sharply (0.5B drops from 29% to 10%).
RQ4	Entity load instances generated; model evaluation output pending.
RQ5	Structural operations display high variance; MERGE and UNDO reach 92% under 3B CoT, while SWAP remains low (18%).

Chapter 8

Ethical, Social, and Environmental Considerations

8.1 Data Integrity and Research Ethics

DWS-Bench strictly generates synthetic evaluation instances from deterministic symbolic traces. Symbolic labels are computed independently of natural-language text generation, preventing label hallucination. Evaluation protocols report all benchmark slices without selectively filtering low-performing conditions.

8.2 Social Accessibility and Environmental Impact

Small Language Models enable accessible deployment on localized edge devices. Evaluating efficiency vs. reasoning accuracy assists resource-constrained institutions. While model evaluations consume power, evaluating sub-10B parameter models requires significantly less compute than frontier multi-billion parameter evaluations.

Chapter 9

Challenges and Limitations

1. **Residual Confounding:** Length-matching (± 3 words) reduces length confounding but does not eliminate all token-level variances.
2. **Synthetic Domain Transfer:** Synthetic text provides clean experimental control but lacks natural language ambiguity.
3. **Operation Vocabulary Limits:** The eight operations do not cover spatial topology or containment relations.
4. **Model Architecture Scope:** Evaluation focuses on Qwen2.5 instruction-tuned models; results may vary for other architecture families.

Chapter 10

Conclusion and Future Work

10.1 Conclusion

This report introduced **DWS-Bench**, a controlled benchmark framework isolating dynamic world-state reasoning in Small Language Models. By decoupling symbolic world state updates from natural language rendering, DWS-Bench provides deterministic ground truth across eight canonical operations. Evaluation of Qwen2.5 models revealed that reasoning performance is highly non-monotonic, and that high final-answer accuracy (49.91% in 3B CoT) can mask severe intermediate trajectory failure (2.82% step-wise accuracy).

10.2 Future Work

Future extensions include:

1. Broader query types and multi-object questions;
2. Containment, ownership, and additional relational state representations;
3. Richer multi-hop and multi-entity reasoning;
4. Longer and more varied dependency structures;
5. Broader model and parameter-scale coverage;
6. Representation-level and mechanistic analysis of state tracking;
7. Additional naturalistic datasets and transfer settings;
8. Improved statistical controls for residual context-length confounding;
9. Expanded trajectory families and operation semantics.

Appendices

Appendix A

Weekly Meeting Summaries

Table A.1 records weekly progress, decisions, and planned work, reconstructed from the team’s WhatsApp group log (“Batch 22 – Thesis group”).

Table A.1: Weekly meeting summaries and project progress.

No.	Date	Progress / Contribution	Decision / Discussion	Next Plan
1	14–18 May 2026	Started literature review on LLM interpretability and concept representations. Set up the thesis GitHub repository.	Discussed LRH, CRH, steering vectors, and their limitations.	Read background papers and prepare for supervisor discussion.
2	20–22 May 2026	Reviewed additional papers shared by the supervisor.	Discussed the need to understand the papers before selecting a research direction.	Review the remaining papers and identify possible research gaps.
3	28 May–02 June 2026	Continued literature review; supervisor shared work on LLM working memory.	Working memory and reasoning became potential areas of interest.	Continue literature review and arrange the next meeting.

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No.	Date	Progress / Contribution	Decision / Discussion	Next Plan
4	03–05 June 2026	Coordinated and attended a supervisor meeting.	Discussed the team’s progress and ongoing literature exploration.	Continue narrowing down the research direction.
5	10–12 June 2026	Progress was temporarily limited due to ML coursework and quizzes.	Meeting was postponed with supervisor approval.	Resume thesis work after academic deadlines.
6	14–17 June 2026	Prepared to discuss accumulated literature findings with the supervisor.	Meeting was arranged to review progress and possible research directions.	Present findings and identify a feasible research problem.
7	14–15 July 2026	Converged toward SLM reasoning and state-tracking evaluation. Draft proposal was shared.	Identified a possible gap in multi-parameter evaluation of SLM state tracking. Proposed adapting Kim & Schuster’s (ACL 2023) entity-tracking generator.	Refine the research gap and benchmark design.
8	17–23 July 2026	Continued refining the SLM state-tracking proposal after a postponed meeting.	Supervisor meeting was rescheduled.	Discuss the benchmark and research gap in detail.
9	23–31 July 2026	Refined the project toward a controlled dynamic world-state benchmark.	Discussed using a deterministic simulator and controllable task parameters.	Formalize the variables and begin implementation.
10	03 August 2026	Continued literature review and benchmark methodology development.	Focused on isolating different sources of difficulty in SLM state tracking.	Implement the deterministic state-generation and validation pipeline.

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No.	Date	Progress / Contribution	Decision / Discussion	Next Plan
11	09–14 August 2026	Formalized the symbolic-simulator-first approach and experimental variables.	Ground truth should be generated independently from natural-language rendering.	Test alternative rendering approaches on a small pilot set.
12	15 August 2026	Discussed deterministic question generation versus LLM-based paraphrasing. Reviewed related task formulations such as LinearWorld, HandSwap, and Lights.	Supervisor recommended testing both approaches at small scale. Team considered deterministic variation preferable due to lower risk of ground-truth drift.	Run a small comparison and finalize the rendering strategy.

Appendix B

Implementation and Artifacts

B.1 Repository Architecture

Table B.1 lists key repository modules.

Table B.1: Repository structure overview.

Component	Module Scope
Contract	Specification of QuerySpecs and complexity rules
Generator	Candidate symbolic trajectory generation
Validator	Admissibility and ground-truth verification
Renderer	Natural language narrative translation
Evaluator	Model inference and metric computation

B.2 Technical Contribution Matrix

Table B.2 details individual contributions.

Table B.2: Technical contribution matrix.

Member	PR ID	Technical Contribution
MD. Mahiul Kabir	–	Symbolic World Simulator and Experimental Contract implementation.
MD. Mahiul Kabir	–	Repository artifact documentation and implementation validation.
Rahatut Tahrim Mounota	–	Trajectory Generator and Validation Logic implementation.
Rahatut Tahrim Mounota	–	Benchmark validation and result-aggregation support.
Nayeemul Hasan Prince	–	Natural-Language Renderer and Prompt Generation.
Nayeemul Hasan Prince	–	Report synthesis and evidence-alignment review.

B.3 GitHub Metrics and Verification

Tables B.3 and B.4 document repository activity.

Table B.3: GitHub contribution metrics.

Member	Commits	Additions	Deletions
MD. Mahiul Kabir	2	2,728	26
Rahatut Tahrim Mounota	13	787,654	6,457
Nayeemul Hasan Prince	2	6,567	6,823
Total	17	796,949	13,306

Table B.4: Pull Request verification matrix.

Member	PR 1	PR 2
MD. Mahiul Kabir	–	–
Rahatut Tahrim Mounota	–	–
Nayeemul Hasan Prince	–	–

Appendix C

Final Compliance Audit

Table C.1 completes the final submission compliance checklist.

Table C.1: Final DWS-Bench compliance audit checklist.

State	Requirement	Status
Done	Chapters 1–10 contain complete methodological and empirical content.	Completed
Done	Phase 1 generator validation and factor-distribution analysis completed.	Completed
Done	Realized factor associations have been inspected before benchmark freezing.	Completed
Done	Frozen benchmark datasets are used for Phase 2 evaluation.	Completed
Done	Three target SLMs have been evaluated under the fixed protocol.	Completed
Done	Final-answer and step-wise accuracy are reported.	Completed
Done	RQ1–RQ3 results and interpretations are reported.	Completed
Open	RQ4 evaluation remains pending.	Pending
Done	RQ5 structural-operation families are generated, validated, and evaluated within the core benchmark.	Completed
Done	Performance-degradation and error-dynamics analyses are reported.	Completed
Open	Naturalistic reference audit is documented as a secondary analysis.	Pilot / Descriptive
Done	Symbolic ground truth is independent of the natural-language renderer.	Completed
Done	Validator performs the mandatory structural and state-consistency checks.	Completed
Done	Appendix A contains weekly meeting summaries.	Completed
Open	Appendix B contains project and GitHub artifacts.	Pending final artifact verification
Open	GitHub contribution evidence is included where required.	Pending final artifact verification
Done	Chapter 7 contains findings linked explicitly to each research question.	Completed

C.1 Final Presentation Requirements

The final presentation should explicitly communicate:

1. The DWS-Bench architecture;
2. The QuerySpec and experimental contract;
3. The eight-operation capability taxonomy;
4. Phase 1 generator-validation findings;
5. RQ1–RQ5 results;
6. Performance-degradation and error-dynamics findings;
7. The naturalistic reference audit;
8. Major research findings;
9. Limitations;
10. Future extensions.

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